BEACON

Broadscale Evacuation Agentic Coordination, Outreach & Navigation

> Fast, language-inclusive evacuation guidance
> that routes people away from active fire perimeters

Equitable Wildfire Collective

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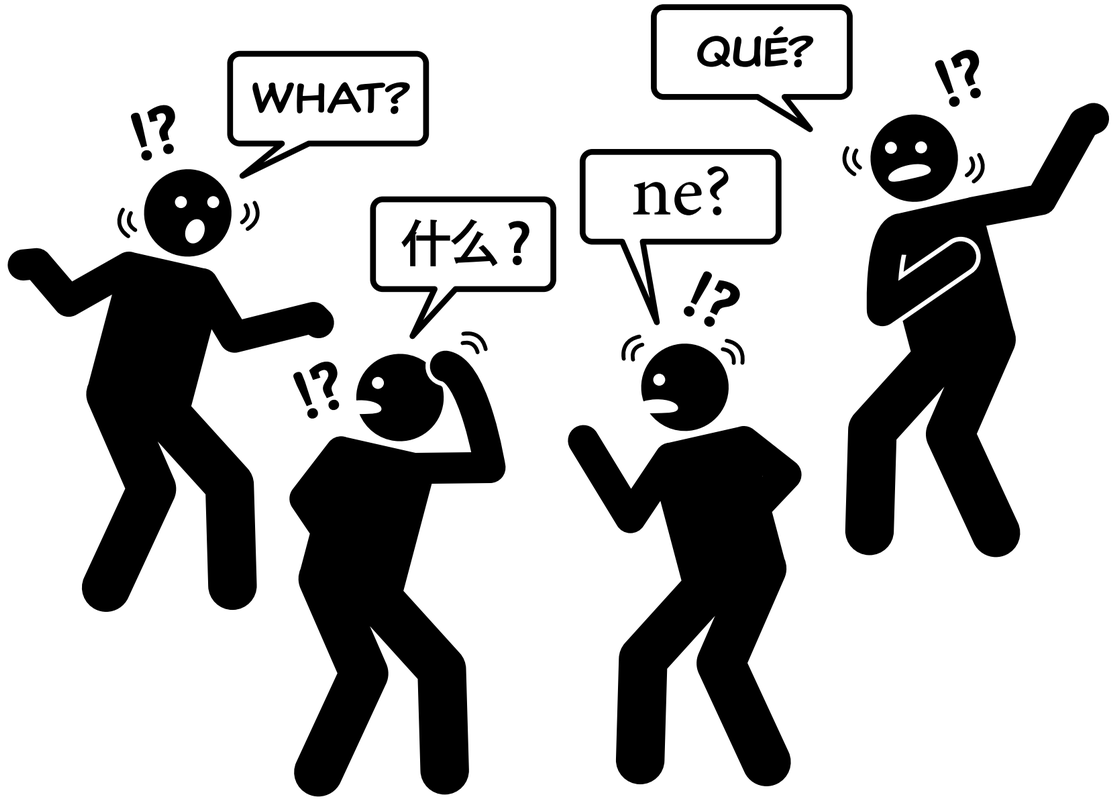
The "English" Assumption



CURRENT SYSTEMIC BARRIERS:

- Fire seasons are 84 days longer than in the 1970s and the evacuation alerts are still English-only
- 26M U.S residents have limited English proficiency (LEP) with 9.5M speaking neither English nor Spanish
- 80% of emergency alerts are English-only
- Multilingual alerts aren't federally required until 2028

TARGET USER:

Non-English speaking residents in wildfire-prone areas

The Data

PRIMARY SOURCE

WatchDuty

- Active fire polygons with perimeter boundaries and timestamps
- Used for real time danger assessment & ORS polygon-avoidant route

WEATHER MODEL

HRRR (NOAA)

- Hourly wind speed, direction, humidity at 3 km resolution
- Drives High Wind Potential (HWP) zone calculation

LIVE INPUT

User GPS

- Live coordinates captured at session start
- Haversine distance calculation to the nearest fire perimeter

STATIC DATASET

Shelter & Facility Lists

- Evacuation centers, staging sites, care facilities pulled from OverPass API's tagged facilities list
- No live availability (known limitation)

The Danger Assessment

SPATIAL **User GPS → nearest fire perimeter**

TEMPORAL **Per session, refreshed on open**

BASELINE **Distance only, no wind adj.**

User GPS
Live coordinates at session start

→

WatchDuty Polygons
Active fire perimeters + polygons

→

Haversine Distance
User → newest perimeter (km)

→

Danger Bin
LOW/MEDIUM/HIGH/CRITICAL

→

Agent Response
Tone + checklist severity

LOW
General preparedness tips

MEDIUM
Prepare a go bag, stay alert

HIGH
Evacuate now, evacuation zones nearby

CRITICAL
Call 911, no safe routes available

HWP LAYER | HRRR wind speed, direction & humidity overlaid on distance score; adjusts bin upward when wind is blowing toward user. Refresh cadence driven by fire activity patterns.

The Route Creation

User Location
GPS coordinates

+

Fire Polygons
WatchDuty perimeters

↓

OpenRouteService (ORS)
Polygon-avoidant routing engine

↓

GeoJSON Route
Avoids all active fire perimeters

→

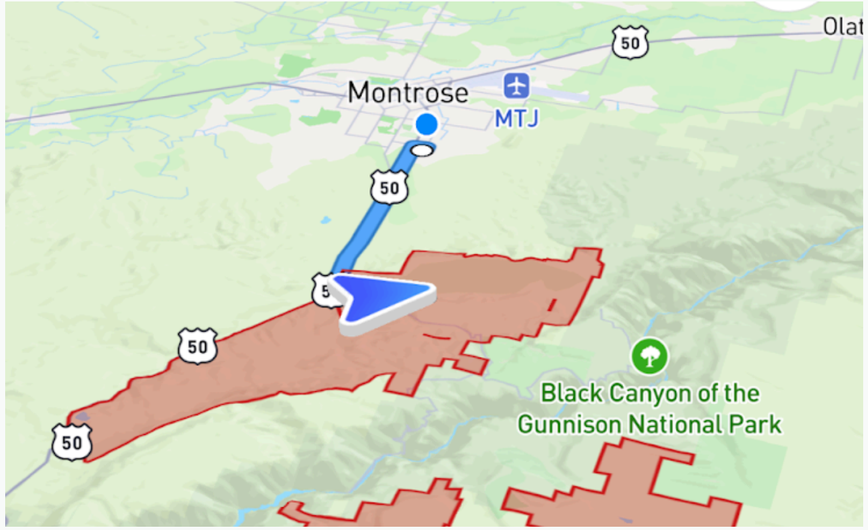
Mapbox
Turn-by-turn navigation

WHY NOT GOOGLE MAPS

Evaluated Google Maps but it repeatedly ignored polygon constraints and routed through active fire perimeters. ORS gives us direct control over avoidance zones.

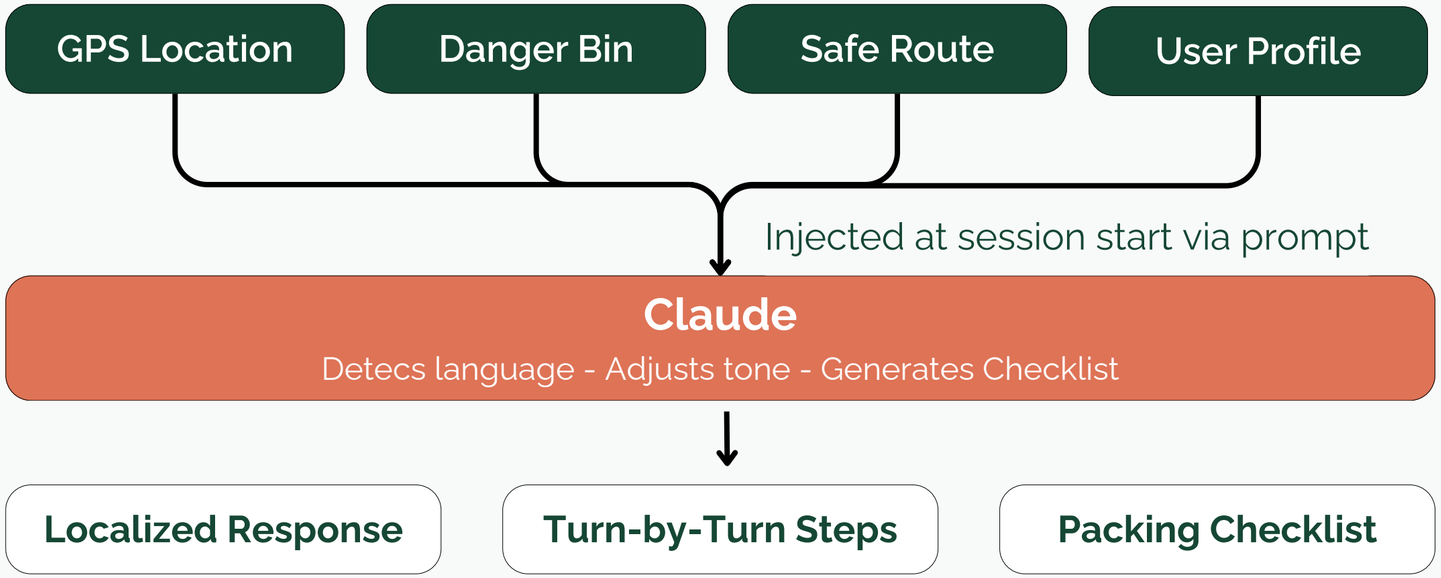
GRACEFUL DEGRADATION

No safe route found	Instruct user to call 911
Fire feed down	Fall back to mock polygon data
Chat context missing	Safe default response & appropriate escalation



The Agent

AGENT OPERATING MODEL



LANGUAGE DETECTION

- Single system prompt instruction
- Anchored to most recent message so language can shift mid-conversation without breaking

TONE BY DANGER BIN

- LOW → calm, general preparedness tips
- MEDIUM → prepare a go bag, stay alert
- HIGH → active evacuation zones nearby, evacuate now
- CRITICAL → no safe route or exits blocked, call emergency services

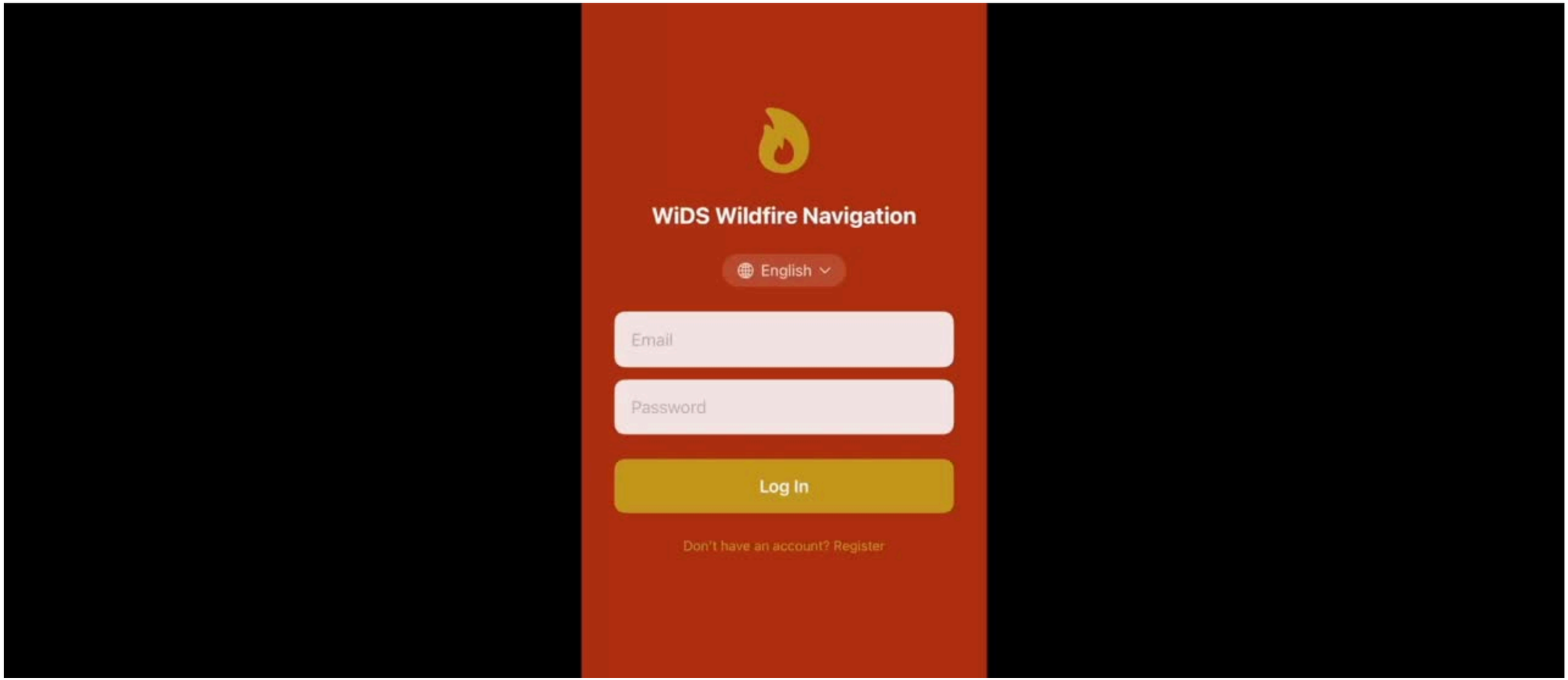
GOVERNANCE

- Operates autonomously within defined safety rules
- No hallucinated shelter info
- Escalates to emergency services when no safe route; primary safeguard at system limits

LANGUAGE DETECTION

- Summarization fires when history exceeds threshold
- Compresses into synthetic exchanges while preserving safety context and avoiding token limits

The Application in Action



The Results

What Works

- Language detection
- Polygon-avoidant routing via ORS
- Context injection at session start
- Graceful degradation

Not yet solved

- No live shelter availability
- High Resolution Rapid Refresh (HRRR) fixed polling doesn't scale
- No human oversight in the loop
- Edge case when no safe route exists, only recourse is emergency services

The Pivot

01

DATA SOURCE

WatchDuty as the Primary Data Source

→ Started with NASA FIRMS satellite hotspots as primary fire detection source

→ WatchDuty's real time data perimeter data proved to be more actionable than satellite hotspots

02

ARCHITECTURE

System Change

→ Original plan was a vulnerability-aware dashboard that focused on showing impacted and underresourced communities

→ Pivoted to a project that was an active way to support a vulnerable community

03

TARGET USER

Population vulnerability

→ Initially focused on rural communities and elderly/disabled populations but face limititations with the data

→ Realized existing tools were not supporting folks whose primary language is not English

The Next Steps

KNOWN CONSTRAINTS

- Requires internet connection and GPS access
- Dependent on WatchDuty API and ORS availability

DATA DEPENDENCIES

- WatchDuty polygon quality and freshness
- HRRR availability for wind-adjusted danger assessment

ASSUMPTIONS

- Users have smartphones with reliable GPS
- WatchDuty perimeters reflect ground truth

WHAT'S TO COME:

- **Live evacuation sources**
- **Pattern-based HRRR refresh**
- **Human review for critical translations**
- **Speech-to-Text**

Thank You!